참고문헌 원문 확보 현황

건설현장 YOLO-LLM 안전관제 시스템 개발 및 실증 연구: 도입 현장과 미도입 현장의 안전관리 실효성 비교를 중심으로

경기대학교 공학대학원 건축·안전공학전공 윤혁 · 지도교수 박종용 · 작성 2026. 11. 8. · 기준: 학위논문 본문 참고문헌 54건

45건	1건	6건	2건
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1. 국내문헌					
국내 01	고용노동부(2025). 2024 산업재해 현황분석	확보	PDF 통계 보고서	—	4
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국내 04	김민기·박성호(2026). 서울시 재난안전관리 지원 생성형 AI 구축과 활용 방안	확보	PDF 연구보고서	—	7
국내 05	김석기(2026). 소규모 건설현장 사망사고의 시스템적 원인 규명을 위한 BERTopic-LLM과 AcciMap 통합 분석 프레임워크 개발	확보	PDF 박사학위논문	—	8
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국내 08	김현수(2026). 통합관제센터 운영을 위한 Vision AI–VLM 비교 및 하이브리드 연구	확보	PDF 석사학위논문	—	11
국내 09	대한민국 정부(2025). 이재명정부 123대 국정과제	확보	PDF 정부 간행물	—	12
국내 10	류수영(2024). CCTV 기반 산업안전 관제 시스템에 관한 연구	확보	PDF 박사학위논문	—	13
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국내 12	맹인영 외(2021). 건설 현장 스마트 안전 장비 도입에 따른 실효성 제고 방안 연구. 안전문화연구, 14	확보	PDF 공개 저자원고	doi.org/10.52902/kjsc.2021.14.47	15
국내 13	박종학(2025). YOLO 모델별 건설 현장 위험 상태 및 객체 인식 성능 비교	확보	PDF 석사학위논문	—	16
국내 14	윤영석(2026). 중·소규모 건설현장 안전관리자 인건비 직접공사비 계상과 안전관리 작동성에 관한 연구	확보	PDF 석사학위논문	https://www.riss.kr/link?id=T17395920	17
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국내 16	이동건(2024). 통합기술수용모델(UTAUT)을 활용한 건설근로자들의 스마트안전장비 수용에 영향을 미치는 요인에 관한 연구	확보	PDF 석사학위논문	—	19
국내 17	이석희·고대식·송석일(2025). 건설 안전 모니터링을 위한 장면 그래프와 템플릿 기반 실시간 비디오 캡셔닝 시스템 구현. Journal of KIIT, 23(11)	확보	PDF 학술지 출판본	doi.org/10.14801/jkiit.2025.23.11.279	20
국내 18	이영미(2020). 시각장애학생의 인지 및 적응행동 특성	확보	PDF 박사학위논문	—	21
국내 19	이준호(2024). 건설현장 중대재해 감소를 위한 스마트 안전장비의 지원방법 개선에 관한 연구	확보	PDF 석사학위논문	—	22
국내 20	이창남(2025). 스마트장비(Drone·AIoT)를 활용한 건설현장 안전 관리 활성화 방안에 관한 연구	확보	PDF 박사학위논문	—	23
국내 21	이형도(2024). 객체탐지 알고리즘을 이용한 건설현장 근로자 안전모 실시간 모니터링 기술 활성화 방안 연구	확보	PDF 박사학위논문	—	24
국내 22	한국인터넷진흥원(2026). 영상정보 분석 성능시험 제도 안내서	확보	PDF 안내서	https://www.ksecurity.or.kr/user/bbs/kisis/28/97/bbsDataView/37569.do	25

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국내 23	한소은(2026). 건설현장 '일일 위험성 평가서' 자동 생성 모델: 'Vision-Language Model (VLM)'을 활용하여	확보	PDF 석사학위논문	—	26
국내 24	황병복(2026). 철도 건설사업에서 스마트 건설기술의 도입 및 활용이 안전 성과에 미치는 영향	확보	PDF 박사학위논문	—	27
2. 국외문헌					
국외 01	Bloom, H. S. (1995). Minimum Detectable Effects. Evaluation Review, 19(5)	확보	PDF 학술지 출판본	doi.org/10.1177/0193841X9501900504	28
국외 02	Brown, T. et al. (2020). Language Models are Few-Shot Learners. NeurIPS 33	확보	PDF 학술대회 논문	https://proceedings.neurips.cc/paper/2020/hash/1457c0d6bfcb4967418bfb8ac142f64a-Abstract.html	29
국외 03	Chen, X., & Zou, Z. (2026). Are large pre-trained vision language models effective construction safety inspectors. Data-Centric Engineering, 7	확보	PDF 학술지 출판본	doi.org/10.1017/dce.2026.10044	30
국외 04	Chong, H.-Y. et al. (2023). The Adoption Intentions of Wearable Technology for Construction Safety. Buildings, 13(11)	확보	PDF 학술지 출판본	doi.org/10.3390/buildings13112747	31
국외 05	Cochran, W. G. (1969). The Use of Covariance in Observational Studies. Applied Statistics, 18(3)	미확보	원문 PDF 미보유 — Crossref·출판사 서지와 초록으로 존재 확인	doi.org/10.2307/2346587	—
국외 06	Costello, A. B., & Osborne, J. W. (2005). Best practices in exploratory factor analysis. PARE, 10(7)	확보	PDF 학술지 출판본	doi.org/10.7275/jyj1-4868	32
국외 07	Cvach, M. (2012). Monitor alarm fatigue: An integrative review. Biomedical Instrumentation & Technology, 46(4)	확보	PDF 학술지 출판본	—	33
국외 08	Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. MIS Quarterly, 13(3)	확보	PDF 학술지(JSTOR)	https://www.jstor.org/stable/249008	34
국외 09	Green, D. M., & Swets, J. A. (1966). Signal Detection Theory and Psychophysics. Wiley	미확보	단행본 원전 미보유 — 보유 문헌(국외 13·17)의 인용으로 재확인	ISBN 9780471324201	—
국외 10	He, C. et al. (2023). Effects of Demographic Characteristics on Safety Climate and Construction Worker Safety Behavior. Sustainability, 15(14)	확보	PDF 학술지 출판본	doi.org/10.3390/su151410985	35
국외 11	Kim, J. et al. (2025). An Integrated YOLO and VLM System for Fire Detection in Enclosed Environments. ICLR 2025 Workshop, PMLR 296	확보	PDF 학술대회 논문	https://proceedings.mlr.press/v296/kim25a.html	36
국외 12	Kim, Y.-S. et al. (2022). Effectiveness Analysis for Smart Construction Safety Technology (SCST). IJERPH, 19(9)	확보	PDF 학술지 출판본	doi.org/10.3390/ijerph19095203	37
국외 13	Lynn, S. K., & Barrett, L. F. (2014). "Utilizing" signal detection theory. Psychological Science, 25(9)	확보	PDF 학술지 (OnlineFirst판)	doi.org/10.1177/0956797614541991	38
국외 14	Manzey, D. et al. (2014). Decision-making and response strategies in interaction with alarms. Ergonomics, 57(12)	확보	PDF 저자 최종원고	doi.org/10.1080/00140139.2014.957732	39
국외 15	Mi, Y. (2025). Deep Learning-Based Safety Analysis for High-Altitude Construction Workers	확보	PDF 박사학위논문	—	40
국외 16	Pisu, A. et al. (2024). Enhancing workplace safety: A flexible approach for PPE monitoring. Expert Systems with Applications, 238	확보	PDF 학술지 출판본	—	41
국외 17	Stanislaw, H., & Todorov, N. (1999). Calculation of signal detection theory measures. BRMIC, 31(1)	확보	PDF 학술지 출판본	doi.org/10.3758/BF03207704	42
국외 18	Tavakol, M., & Dennick, R. (2011). Making sense of Cronbach's alpha. IJME, 2	확보	PDF 학술지 출판본	doi.org/10.5116/ijme.4dfb.8dfd	43
국외 19	Teizer, J. (2016). The Role of Automation in Right-time Construction Safety. ISARC 2016	확보	PDF 학술대회 논문	doi.org/10.22260/ISARC2016/0024	44

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국외 20	The jamovi project (2026). jamovi (Version 2.7) [Computer Software]	**웹 열람**	소프트웨어(공식 사이트)	https://www.jamovi.org	—
국외 21	Wilson, E. B. (1927). Probable Inference, the Law of Succession, and Statistical Inference. JASA, 22(158)	**확보**	PDF 학술지(JSTOR 스캔본)	doi.org/10.1080/01621459.1927.10502953	45
국외 22	Wu, J. et al. (2025). MonitorVLM: A Vision–Language Framework for Safety Violation Detection in Mining Operations. arXiv:2510.03666v1	**확보**	PDF 프리프린트 (arXiv v1)	https://arxiv.org/abs/2510.03666v1	46
국외 23	Zhang, G. et al. (2026). ConstructAI: From Real-Time Safety Insight to Skill Growth in Deployed Construction AI Systems. AAAI 40(47)	**확보**	PDF 학술대회 논문	doi.org/10.1609/aaai.v40i47.41452	47
3. 법령					
법령 01	「건설기술 진흥법」 제39조·제49조 [법률 제21065호]	**웹 열람**	국가법령정보센터 조문	https://www.law.go.kr/LSW/lsLinkCommonInfo.do?chrClsCd=010202&lsJoLnkSeq=1033555451	—
법령 02	「건설산업기본법」 제40조 [법률 제21034호]	**웹 열람**	국가법령정보센터 조문	https://www.law.go.kr/법령/건설산업기본법/제40조	—
법령 03	「산업안전보건법」 제15·16·17·63·67조 [법률 제22122호, 2026. 10. 27.]	**확보(구판)**	PDF 법령 전문 (2025. 10. 1. 시행 법률 제21065호 판) + 현행본 웹 열람	https://www.law.go.kr/법령/산업안전보건법	48
4. 표준·기술자료					
기술 01	Bosch Security Systems (2024). Alarm Verification for Alarm Management. Application note F01U411472	**확보**	PDF 기술문서	—	49
기술 02	OSHA (연도 미상). Hazard Prevention and Control. Safety and Health Programs	**웹 열람**	웹 문서	https://www.osha.gov/safety-management/hazard-prevention	—
기술 03	Ultralytics (2026a). YOLO (Version 8.4.23) [Computer software]	**웹 열람**	소프트웨어 패키지	—	—
기술 04	Ultralytics (2026b). YOLO26. Ultralytics Docs	**웹 열람**	웹 문서	https://docs.ultralytics.com/models/yolo26/	—

· 번호는 본문 참고문헌 장의 구분별 번호이다. '수록 쪽'은 이 PDF 안에서 해당 원문 첫 쪽이 실린 쪽이다.

· **확보**: 원문 PDF 전체를 보관하고 있으며, 뒤쪽에 서지가 보이는 첫 쪽을 실었다. 학위논문은 원문 1쪽이 dCollection 저작권 이용허락 안내이므로 표지(원문 2쪽)를 실었다.

· **확보(구판)**: 산업안전보건법은 2025. 10. 1. 시행본(법률 제21065호) PDF를 보관하고, 인용한 현행본(법률 제22122호)은 국가법령정보센터에서 조문을 대조하였다(인용한 다섯 조문의 제목·요지는 두 판본에서 같음).

· **웹 열람**: 파일로 내려받는 문헌이 아니라 법령 조문·소프트웨어·웹 문서이므로 표의 주소에서 확인한다.

· **미확보**: Cochran(1969)은 원문 PDF를 확보하지 못하고 Crossref와 출판사 서지·초록으로 존재와 서지를 확인하였다. Green & Swets(1966)는 단행본 원전을 보유하지 않으며, 보유 문헌인 Lynn & Barrett(2014)와 Stanislaw & Todorov(1999)에서 이 원전을 인용한 부분을 확인하였다.

· 판본 주: 국내 12는 공개 저자원고, 국외 13은 OnlineFirst판, 국외 14는 저자 최종원고, 국외 22는 arXiv v1이며 본문 인용 쪽수는 보관 판본 기준이다.

산업재해조사 개요

1. 조사 목적

이 조사는 산업재해의 산업별, 규모별, 지역별, 발생시기별, 원인별 분포와 재해 근로자의 성별, 연령별, 근속기간별 등 특성을 파악하여 산업재해예방 정책 수립의 기초자료로 제공하는 데 그 목적이 있다.

2. 법적 근거

통계법 제18조에 따른 일반통계(승인번호 118006호*)이다.

* 11806호 → 118006호(2016. 8. 28. 승인번호 체계변경, 통계청)

3. 조사 연혁 및 주기

1975. 8. 11. 총리령 제154호에 따라 보고양식 44-4호로 최초 승인을 받은 이후 매년 조사를 실시하고 있다.

4. 조사 대상

산업재해보상보험법 적용사업체에서 발생한 산업재해 중 산업재해보상보험법에 의한 업무상 사고 및 질병으로 승인을 받은 사망 또는 4일 이상 요양을 요하는 재해(지방고용노동관서 산재미보고 적발사망 포함)를 조사대상으로 하였다.

※ 공무원연금법, 사립학교교직원연금법, 군인연금법 적용 사업장의 업무상재해 발생현황은 별도로 수록하였음

※ 어선원재해보상보험 및 어업인안전보험, 농업인안전보험 적용 사업장의 업무상재해 발생현황을 별도로 수록하였음

고용노동부 보도자료

보도시점 2026. 3. 31.(화) 12:00 (2026. 4. 1.(수) 조간)

2025년 유족급여 승인 기준 사고사망 재해 현황

고용노동부(장관 김영훈)는 산재보상 통계에 기반한 **'2025년 유족급여 승인 기준 사고사망 재해 현황'**을 발표했다. 이는 2025년 한 해 동안 근로복지공단에서 유족급여가 승인된 건수를 집계하여 산출한 통계*로 **2025년 유족급여가 승인된 사고사망자는 872명**('24년 827명, +45명)이었다.

* '25년 이전에 발생한 사고가 '25년에 유족급여 승인을 받는 경우에는 '25년 통계에 반영

사고사망만인율은 산재보험 적용 확대 등으로 **가입자 수가 증가**함에 따라 **0.38‰**(0.383‰)로 **전년**(0.386‰)과 **비슷한 수준**이었다.

주요 내용은 다음과 같다.

업종별로는 ▲**건설업 361명**(41.4%, +33명), ▲**운수창고통신업 176명**(20.2%, +38명), ▲**제조업 164명**(18.8%, △23명), ▲**기타의 사업**(서비스업) **140명**(16.1%, △5명) 순이었다.

규모별로는 ▲**5인 미만 354명**(40.6%, +45명), ▲**5~49인 332명**(38.1%, △29명), ▲**50~299인 121명**(13.9%, +11명), ▲**300인 이상 65명**(7.5%, +18명) 순으로 나타났다.

유형별로는 ▲**떨어짐 280명**(32.1%, +2명), ▲**사업장 외 교통사고 123명**(14.1%, +36명), ▲**부딪힘 81명**(9.3%, +1명), ▲**깔림·뒤집힘 69명**(7.9%, +15명) 순이었다.

종사자별로는 ▲**근로자 730명**(83.7%, +5명), ▲**노무제공자 137명**(15.7%, +36명), ▲**중소기업 사업주 5명**(0.6%, +4명) 순이었다. 이 중 노무제공자는 **전속성 요건 폐지로 가입 대상이 확대**되면서 유족급여 승인이 증가한 영향이 있었으며, 유형별로는 **사업장 외 교통사고 사망자가 96명**(70.1%, +34명)으로 가장 많았다.

외국인 사고사망자는 **77명**으로 **전년(102명) 대비 25명이 감소**하였는데, 이는 화성 화재 사고('24.6.24.)의 영향으로 전년도 외국인 사고사망자가 크게 증가했던 점이 올해 증감에 일부 반영된 것으로 보인다.

※ 보다 자세한 통계자료는 고용노동부 홈페이지(www.moel.go.kr) → 정책소개 → 정책자료실 → 2025년 산업재해 현황에서 확인하실 수 있습니다.

CS-24-E6-002



스마트 안전장비 활용 가이드라인

국토교통부 국토안전관리원

2024.03

서울시 재난안전관리 지원 생성형 AI 구축과 활용 방안

김민기 박성호

서울연구원
The Seoul Institute

工學博士學位論文

소규모 건설현장 사망사고의 시스템적 원인 규명을 위한 BERTopic-LLM과 AcciMap 통합 분석 프레임워크 개발

Development of Integrated Analysis Framework Using BERTopic-LLM and AcciMap for Identifying Systemic Causes of Fatal Accidents in Small-Scale Construction Sites

충 북 대 학 교 대 학 원

방재공학협동과정 재난관리전공

김 석 기

2026년 2월

Collection @ chungbuk

중·소규모 건설현장 재해예방을 위한 스마트 건설안전기술 활용 방안 2022 김용선

박사학위논문

중·소규모 건설현장 재해예방을 위한 스마트 건설안전기술 활용 방안

Smart Construction Safety Technology Utilization for Disaster Prevention at Small and Medium Construction Sites

2022년 8월

인천대학교 대학원

안전공학과

김 용 선

博士學位 請求論文
指導敎授 朴 承 熙

건설현장 위험상황 이해 및 대응을 위한 인공지능 안전인지 프레임워크 연구

成均館大學校 一般大學院
글로벌스마트시티融合專攻
災難管理專攻
金 兌 憲

SUNG KYUN KWAN UNIVERSITY

제 144 회 석사학위 논문
지도교수 박인선 박호현

통합관제센터 운영을 위한 Vision AI - VLM 비교 및 하이브리드 연구

A Comparative Study of Vision AI Model and VLM with Hybrid Approach for Integrated Control Center Operations

중앙대학교 대학원
ICT안전학과 ICT안전전공
김 현 수

2026년 2월

이재명정부
123대 국정과제

2025년 9월

대한민국 정부

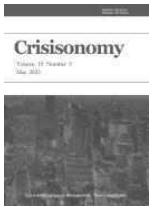
박사학위논문

CCTV 기반 산업안전 관제 시스템에 관한 연구

A Study on CCTV-based Occupational Safety Control System

2023년 12월

배재대학교 대학원
컴퓨터공학과 컴퓨터공학전공
류 수 영



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Crisisonomy

An Analysis of the Performance and Implementation Effects of Smart Safety Equipment at Construction Sites

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Abstract

This study aims to empirically analyze the risk detection accuracy and accident reduction effect of smart safety equipment at small and medium sized construction sites, and to suggest policy directions for government support. Using data from 13 sites equipped with AI-based CCTV in 2023, we evaluated the detection accuracy of safety helmet violations, falls, and fire occurrences based on the YoLo v4 algorithm. Results showed that fire and helmet-related risks exceeded the acceptable threshold of 90.0%, while fall detection was below it. Although the sample size was limited, sites with smart equipment recorded a 0.0% accident rate, compared to 0.461% at general small sites, demonstrating a clear preventive effect. These findings confirm the effectiveness of smart safety technologies in field conditions. To enhance accuracy and applicability, adoption of advanced algorithms and improved equipment is necessary. Additionally, the government should provide financial incentives and institutional support to promote broader deployment of smart safety systems.

Key words: smart safety equipment, smart equipment, smart technology, AI CCTV, AI safety equipment

Ⅰ. 서론

건설업에 종사하는 근로자 수는 223만 명으로 전체 산업에 종사하는 근로자 수 2,026만 명의 11.0%에 불과하다. 그러나 최근 2년간 건설업에서 사망자가 2022년 53.0%(341/644명), 2023년 50.7%(303/598명)가 집중적으로 발생하고 있다(Ministry of Land, Infrastucture and Transport, 2024). 특히, 건설업 에서도 50억 원 미만의

Table 1. Death toll at small construction sites

Sortation / Year	Less than 5 billlion won	More than 5 billlion won
2022	226	115
2023	181	122

※ Sources: Ministry of employment and labor 2024. e-National indicators - Industrial accident status, reconstitution.

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www.kci.go.kr

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건설 현장 스마트 안전 장비 도입에 따른 실효성 제고 방안 연구

국 문 요 약

최근 국내 건설산업은 정부정책으로 인해 변화를 요구받고 있으며, 제조업 등에 비해 2배 이상 사망재해가 발생하는 등 안전 문제를 해결하기 위한 필요성이 제기된다. 변화에 대응하기 위해서는 건설기술의 혁신이 필요한 상황이다. 현재 국내 건설 현장은 4차 산업 현명을 맞이하여 BIM, 드론, ICT 기반 현장관리 기술 등 스마트 건설기술이 일부 도입되어 활용되거나 시범적으로 적용되고 있으며, 일부 스마트 기술들은 아직 연구개발 중으로 전면 활용이 어렵지만, 학계, 발주처 및 건설업체 등에서는 활용방안을 모색하는 등 현장 적용성에 관해 연구를 진행 중이다. 2020년부터 건설사 및 공공기관은 IoT, 센서, 웨어러블 등의 스마트 안전 장비 도입을 본격화하고 있지만, 국내 건설 현장은 전반적으로 스마트 안전관리 체계 구축이 시급하며, 따라서 「스마트 건설안전 시스템」 모델 개발을 통해 효율적인 안전관리 시스템 구축이 필요하다.

본 연구에서는 건설 현장의 스마트 안전 장비 도입에 따른 실효성 제고 방안으로 공공부문 300억이상의 건설 현장 스마트 안전 장비 적용실태 사례조사와 건설 현장 전문가 설문 분석을 통하여 방안을 도출하였으며, AHP 분석법으로 상대 비교를 통한 우선순위를 선정하는 등 결과를 얻을 수 있었다.

연구의 결과로는 건설 현장 스마트 안전 장비 종류를 계층화하고, 공공기관 발주 공사금액 300억이상 10개 현장에 대해 스마트 안전 장비 적용사례, 스마트 안전 장비 비용 적용사례를 1차로 조사한 결과를 바탕으로 건설 현장 전문가들의 설문 분석을 AHP 분석을 통해 어느 정도 신뢰성이 확보되는 것으로 평가하며, 실효성이 높은 스마트 안전 장비를 도출하여 제시함으로써 건설 현장의 스마트 안전 장비 선정에 활용할 수 있으리라 판단된다. 또한, AHP 분석에서도 도출됐듯이 관제 시스템을 통하여 사고당사자인 근로자의 위험 요소를 상시모니터링하여 사고위험 요소를 사전에 제거해야만 사고는 예방될 수 있다고 판단된다. 현장 조사 결과와 설문 분석 결과로 볼 때 건설 현장 스마트 안전 장비 비용에 대한 계상 주체와 계상금액 설계가 표준화되지 못한 부분은 정부에서 기준을 통계방식으로 요율을 표준화하여 발주자가 일괄 적용하고 시공사가 운영하여 효과를 높이는 것이 사고 예방에 실질적으로 도움이 될 것으로 판단된다.

본 연구를 활용하여 건설 현장의 스마트 안전 장비의 실효성 제고 적용을 통한 안전 활동 수준 향상으로

2024학년도
석사학위논문

YOLO(You Only Look Once) 모델별 건설 현장 위험 상태 및 객체 인식 성능 비교

지도교수 : 문 유 미

경기대학교 공학대학원

건축·안전공학전공

박 종 학

2025학년도
석사학위논문

중·소규모 건설현장 안전관리자 인건비 직접공사비 계상과 안전관리 작동성에 관한 연구

지도교수 : 박 종 용

경기대학교 공학대학원

건축·안전공학전공

윤 영 석

명지대학교

박사학위 논문

건설현장의 안전 확보를 위한 딥러닝 기반의 시스템비계 설치 이상감지 연구

A Study on the Deep Learning-Based Integrity Assessment of System Scaffold Installations Ensuring Safety at Construction Sites

명지대학교 대학원

재난안전학학과간협동과정

이 기 수

지도교수 신 동 일

2024년 8월

인하대학교
INHA UNIVERSITY

工學碩士學位 請求論文

통합기술수용모델(UTAUT)을 활용한 건설근로자들의 스마트안전장비 수용에 영향을 미치는 요인에 관한 연구

A Study on the Factors Affecting the Acceptance of Smart Safety Technologies by Construction Workers Using the Unified Theory of Acceptance and Use of Technology (UTAUT) Model

2024년 8월

인하대학교 공학대학원

건축토목공간정보공학과(건축공학전공)

이 동 건

국내문헌 17 ■ 확보
이석희·고대식·송석일(2025). 건설 안전 모니터링을 위한 장면그래프와 템플릿 기반 실시간 비디오 캡셔닝 시스템 구현. Journal of KIIT, 23(11)
보관 파일: 02.reference/본문 반영/17. 건설 안전 모니터링을 위한 장면그래프와 템플릿 기반 실시간 비디오 캡셔닝 시스템 구현.pd.pdf | 원문 1쪽 / 전체 4쪽

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http://dx.doi.org/10.14801/jkiit.2025.23.11.279

건설 안전 모니터링을 위한 장면그래프와 템플릿 기반 실시간 비디오 캡셔닝 시스템 구현

이석희*, 고대식**, 송석일***

Implementation of a Real-Time Video Captioning System based on Scene Graphs and Templates for Construction Safety Monitoring

Seokhee Lee*, Daesik Ko**, and Seokil Song***

본 과제(결과물)는 2025년도 교육부 및 충청북도의 재원으로 충북RISE센터의 지원을 받아 수행된 지역혁신중심대학지원체계(RISE) 글로컬대학30의 결과입니다(2025-RISE-11-004)

요 약

기존 대규모 비전-언어 모델(VLM)과 트랜스포머 기반 비디오 캡셔닝 모델은 높은 성능을 보이지만, 해석 가능성 부족, 환각, 지연 등으로 산업현장 적용에 문제가 있다. 이러한 한계를 극복하기 위해 본 논문에서는 제안하는 캡셔닝 방법은 규칙 및 템플릿 기반의 투명한 추론 경로를 도입하여 모델을 경량화하고, 전 과정(검출 - 장면 그래프 생성 - 템플릿 선택 - 캡셔닝)의 로깅을 통해 설명 가능성과 사후 검증 기능을 보장한다.

Abstract

Large-scale vision–language models and Transformer-based video captioners deliver strong accuracy but remain ill-suited to industrial use due to limited interpretability, hallucination risks, and inference latency. We address these limitations with a lightweight captioning framework that employs a transparent rules-and-templates reasoning path and end-to-end operability logs—spanning detection, scene-graph construction, template selection, and caption generation—to provide rigorous explainability and reliable post-hoc auditability.

Keywords
captioning, scene graph, VLM, video, template

Ⅰ. 서 론

이 논문에서는 2024년 276명의 사고 사망자가 발생한 건설현장의 안전관리를 위하여 CCTV 영상을 사람이 이해할 수 있는 설명문으로 자동 생성하는 이미지/비디오 캡셔닝(Captioning) 기술을 개발한다. 이미지/비디오 캡셔닝 분야에서는 지난 수년간 딥러닝을 활용한 접근법이 주류를 이루어왔다. 특히 트랜스포머(Transformer) 기반의 캡셔닝 모델들은 뛰어난 성능을 보여주며 대규모 데이터 셋에서 SOTA (State-of-The-Art)를 달성해왔다. 그러나 이러한 종단간 딥러닝 모델은 캡셔닝 과정이 블랙박스 형태로

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문학박사 학위논문

시각장애학생의 인지 및 적응행동 특성

특수교육학과 시각장애아교육전공

이 영 미

지도교수 이 해 균

2020년 8월

대구대학교 대학원

석사학위논문

건설현장 중대재해 감소를 위한 스마트 안전장비의 지원방법 개선에 관한 연구

A Study on the Economic Efficiency of Smart Safety Equipment in Construction Site

이 준 호

한양대학교 융합산업대학원

2024년 2월

2024학년도

박사학위논문

스마트장비(Drone·AIoT)를 활용한 건설현장 안전 관리 활성화 방안에 관한 연구

지도교수 : 이 병 길

경기대학교 대학원

건설안전학과

이 창 남

2023학년도
박사학위논문

객체탐지 알고리즘을 이용한 건설현장 근로자 안전모 실시간 모니터링 기술 활성화 방안 연구

지도교수 : 최병정

경기대학교 대학원

건설안전학과

이 형 도

영상정보 분석 성능시험 제도 안내서

2026. 9.



KISA 한국인터넷진흥원

석사학위논문

건설현장 '일일 위험성 평가서' 자동 생성 모델

– 'Vision-Language Model (VLM)'을 활용하여 –

2026 년 02 월

인천대학교 대학원

건축학과

한소은

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■ 확보
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박사학위논문

철도 건설사업에서 스마트 건설기술의 도입 및 활용이 안전 성과에 미치는 영향

The Impact of the Adoption and Utilization of Smart Construction Technology on Safety Performance in Railway Construction Project

2026년 2월

우송대학교 일반대학원

철도공학과

황 병 복

This article describes a simple way to assess the statistical power of experimental designs. The approach presented is based on the concept of a minimum detectable effect, *which, intuitively, is the smallest true impact that an experiment has a good chance of detecting. The article illustrates how to compute minimum detectable effects and how to apply this concept to the assessment of alternative experimental designs. Applications to impact estimators for both continuous and binary outcome measures are considered.*

MINIMUM DETECTABLE EFFECTS

A Simple Way to Report the Statistical Power of Experimental Designs

HOWARD S. BLOOM
New York University

WHAT IS A MINIMUM DETECTABLE EFFECT?

The *minimum detectable effect* of an experiment is the smallest effect that, if true, has an X% chance of producing an impact estimate that is statistically significant at the Y level.

X is the statistical power of the experiment for an alternative hypothesis equal to the minimum detectable effect. Y is the level of statistical significance used to decide whether or not a true effect exists.

For example, consider the minimum detectable effect of an experimental study of a job training program defined as the true positive earnings effect for which the experiment has 80% power, using a one-sided hypothesis test at the .05 significance level. If the minimum detectable effect of hypothetical Design A for the experiment is $500, it has an 80% chance of producing a significant positive impact estimate (indicating that a positive impact exists) when the true impact is $500. If the minimum detectable effect of hypotheti-

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Language Models are Few-Shot Learners

Tom B. Brown* **Benjamin Mann*** **Nick Ryder*** **Melanie Subbiah***

Jared Kaplan† **Prafulla Dhariwal** **Arvind Neelakantan** **Pranav Shyam**

Girish Sastry **Amanda Askell** **Sandhini Agarwal** **Ariel Herbert-Voss**

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Christopher Hesse **Mark Chen** **Eric Sigler** **Mateusz Litwin** **Scott Gray**

Benjamin Chess **Jack Clark** **Christopher Berner**

Sam McCandlish **Alec Radford** **Ilya Sutskever** **Dario Amodei**

Abstract

We demonstrate that scaling up language models greatly improves task-agnostic, few-shot performance, sometimes even becoming competitive with prior state-of-the-art fine-tuning approaches. Specifically, we train GPT-3, an autoregressive language model with 175 billion parameters, 10x more than any previous non-sparse language model, and test its performance in the few-shot setting. For all tasks, GPT-3 is applied without any gradient updates or fine-tuning, with tasks and few-shot demonstrations specified purely via text interaction with the model. GPT-3 achieves strong performance on many NLP datasets, including translation, question-answering, and cloze tasks. We also identify some datasets where GPT-3's few-shot learning still struggles, as well as some datasets where GPT-3 faces methodological issues related to training on large web corpora.

1 Introduction

NLP has shifted from learning task-specific representations and designing task-specific architectures to using task-agnostic pre-training and task-agnostic architectures. This shift has led to substantial progress on many challenging NLP tasks such as reading comprehension, question answering, textual entailment, among others. Even though the architecture and initial representations are now task-agnostic, a final task-specific step remains: fine-tuning on a large dataset of examples to adapt a task agnostic model to perform a desired task.

Recent work [RWC+19] suggested this final step may not be necessary. [RWC+19] demonstrated that a single pretrained language model can be zero-shot transferred to perform standard NLP tasks

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RESEARCH ARTICLE

Are large pre-trained vision language models effective construction safety inspectors

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Keywords: computer vision; construction safety inspection; multi-modal dataset; natural language processing; vision language models

Abstract

Construction safety inspections typically involve a human inspector identifying safety concerns on-site. With the rise of powerful vision language models (VLMs), researchers are exploring their use for tasks such as detecting safety rule violations from on-site images. However, there is a lack of open datasets to comprehensively evaluate and further fine-tune VLMs in construction safety inspection. Current applications of VLMs use small, supervised datasets, limiting their applicability in tasks they are not directly trained for. In this article, we propose the *ConstructionSite 10 k*, featuring 10,000 construction site images with annotations for three inter-connected tasks, including image captioning, safety rule violation visual question answering (VQA), and construction element visual grounding. Our subsequent evaluation of current state-of-the-art large pre-trained VLMs shows notable generalization abilities in zero-shot and few-shot settings, while additional training is needed to make them applicable to actual construction sites. This dataset allows researchers to train and evaluate their own VLMs with new architectures and techniques, providing a valuable benchmark for construction safety inspection.

Impact statement

Vision-language models (VLMs) are a type of AI model that processes images as input and generates textual output. Large pre-trained VLMs, equipped with billions of parameters and trained on vast image-text pairs from the internet, are capable of performing tasks they were never explicitly trained on, using only a few prompts. This flexibility allows such models to be applied to a wide range of construction safety-related tasks without requiring changes to their architecture or additional training. This article introduces an evaluation framework for assessing how well large pre-trained VLMs can perform construction safety tasks, along with a comprehensive dataset specifically designed for this purpose. Practitioners can use this dataset to fine-tune their own VLM to suit specific needs for safety inspection.

1. Introduction

Hazardous working conditions and risky behaviors at construction sites accounted for a concerning 21% of all workplace fatalities in the United States in 2022 (US-BUREAU-OF-LABOR-STATISTICS, n.d.). To ensure compliance with safety regulations, such as wearing proper personal protective equipment

This research article was awarded Open Data badge for transparent practices. See the Data Availability Statement for details.

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buildings

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Article

The Adoption Intentions of Wearable Technology for Construction Safety

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Abstract: Wearable technology (WT) is vital for proactive safety management. However, the adoption and use of WTs are very low when it comes to construction safety. This study proposes a hybrid model, combining elements of the technology acceptance model and the theory of planned behaviour model, with the aim of determining the factors predicting the adoption intention of WTs for construction safety. A mixed-method approach was used to test the model, namely the structural equation model (SEM) and fuzzy-set qualitative comparative analysis (fsQCA). The results show that no single predictor can significantly drive the adoption intention of all six WTs, namely smart wearable sensors, smart safety hats, smart safety vests, smart insoles, smart safety glasses, and smart wristbands, except for the uncovered effective combinations based on each WT individually. This research contributes to new insights into the antecedents of the adoption intention of WTs for construction safety, which are also useful for other technologies.

Keywords: wearable technology; construction safety; adoption intention; mixed method; effective combinations; Australia

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1. Introduction

Construction sites are high-risk environments due to the various environmental risks, high physical demand, and constantly changing site conditions [1–3]. Despite increased efforts to improve the industry's dismal safety record, mortality and injury rates in the global construction industry remain high [4]. Safety management has a reputation for being passive, as it primarily responds to problems after they have occurred [4]. Current safety hazard identification procedures, such as site inspections, incident reports, and ticking off safety checklists, rely heavily on expert opinions to identify dangers [5], making safety management vulnerable to errors. As a result, many academics and practitioners have paid more attention to using emerging technologies in safety-critical systems in order to improve safety and operator performance [6,7].

Wearable technology (WT) is one of the most effective means for proactive safety management. WTs can be described as a category of electronic devices that are either worn directly on the body or incorporated into clothing, fashion accessories, and other everyday items [8], which are also known as "wearables", "wearable devices", and "wearable sensing devices". WTs generally have two functions: a form of communication that allows access to real-time information and the capacity for data input with local storage. Therefore, WTs have the potential to transform conventional practices in construction projects by capturing and managing information, thereby improving safety in the construction industry. As technology adoption is the beginning stage of any business [9], and an emerging WT that is not embraced by its targeted users will never be a success, decision-makers or managers should understand the factors that influence users' decisions to use a particular WT, especially in the early development and diffusion stages. Thus, to achieve the goal of

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Best practices in exploratory factor analysis: four recommendations for getting the most from your analysis

Anna B. Costello

Jason Osborne

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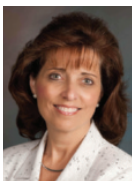
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Monitor Alarm Fatigue
An Integrative Review

Maria Cvach

About the Author



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Abstract

Alarm fatigue is a national problem and the number one medical device technology hazard in 2012. The problem of alarm desensitization is multifaceted and related to a high false alarm rate, poor positive predictive value, lack of alarm standardization, and the number of alarming medical devices in hospitals today. This integrative review synthesizes research and non-research findings published between 1/1/2000 and 10/1/2011 using The Johns Hopkins Nursing Evidence-Based Practice model. Seventy-two articles were included. Research evidence was organized into five main themes: excessive alarms and effects on staff; nurse's response to alarms; alarm sounds and audibility; technology to reduce false alarms; and alarm notification systems. Non-research evidence was divided into two main themes: strategies to reduce alarm desensitization, and alarm priority and notification systems. Evidence-based practice recommendations and gaps in research are summarized.

Medical devices generate enough false alarms to cause a reduction in responding known as the cry wolf effect. Frequent alarms are distracting and interfere with clinicians performing critical tasks and may lead to staff disabling alarm systems.

Background

A cacophony of sound echoes through the modern hospital. Bells, beeps, chimes, and horns are all part of the noise-polluted environment that patients, families, and staff endure. They may be exposed to as many as 700 physiologic monitor alarms per patient per day.[1] The myriad of medical device alarms has created an environment that poses significant risk to patient safety. Device alarms are intended to alert clinicians of a hazardous condition and potential problems. However, when a caregiver is subjected to too many alarms, it disrupts his or her usual workflow and may result in errors due to omission, distraction, or inattention.

The ECRI Institute, a nonprofit organization that uses applied scientific research in healthcare to establish best practices for improving patient care, publishes an annual top ten technology hazards list. "Alarm hazards" is the number one health technology hazard for 2012.[2] Such hazards include inappropriate alarm modification, alarm fatigue, modifying alarms without restoring them to their original settings, and improperly relaying alarm signals to the appropriate person.[3] The problem of excessive alarms resulting in alarm fatigue has been reported in research literature for many years.[4-10] Studies have indicated that the presence of false and/or clinically insignificant alarms ranges from 80%–99%.[4-6]

Medical devices generate enough false alarms to cause a reduction in responding known as the cry wolf effect.[11] Frequent alarms are distracting and interfere with clinicians performing critical tasks and may lead to staff disabling alarm systems. Excessive false positive alarms result in caregiver apathy and desensitization such that real events are less likely to be acted upon.[12,13]

National attention to alarm hazards was spurred in 2010 by the death of a patient at Massachusetts General Hospital that was determined to be due to an alarm that had



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Article

Effects of Demographic Characteristics on Safety Climate and Construction Worker Safety Behavior

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Abstract: Safety climate and safety behavior have been proven to be critical for construction safety. However, few studies have systematically tested the influences of demographic characteristics on construction worker safety, especially in the background of China's construction industry. There is still a gap in the demographics–safety relationship, which may reduce the relevance and effectiveness of safety measures. This study explored the effects of five demographic factors (gender, age, work experience, education level, and employer size) on two safety indicators (safety climate and construction worker safety behavior). The survey data were collected from Chinese construction workers. The *t*-test and one-way ANOVA methods were adopted to examine the demographic differences. The results show that all five demographic factors have significant effects on safety climate, while three (age, education level, and employer size) have significant influences on construction worker safety behavior. However, the remaining two demographic factors (gender and work experience) do not show significant effects on construction worker safety behavior. The results remind the safety management personnel to take a pragmatic approach and to formulate safety measures based on the demographic characteristics of different worker sub-groups.

Keywords: demographic characteristic; safety climate; construction worker safety behavior; sub-group analysis

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1. Introduction

Despite its best efforts, the construction industry continues to be one of the most dangerous industries worldwide [1–3]. Due to rapid urbanization and large-scale infrastructure construction, China's construction industry also suffers from many fatalities and accidents. From 2009 to 2017, 5205 fatal accidents occurred in China's housing construction and municipal engineering projects, resulting in 6354 fatalities [4]. Most of the 55.58 million construction workers in China's 2021 construction industry are peasant workers [5]. They usually have poor living conditions, are separated from their families for a long time, have a low sense of belonging to the local cities, and rarely participate in community activities or trade unions. Of the peasant workers, 38.8% are over 50 years old, and 52.4% possess middle school or below education [6]. These numbers show that there is still much room for China's construction industry to further improve its overall safety performance, and it is urgent to strengthen the safety competence of construction workers because of their general demographic and social characteristics.

Some demographic factors (such as education level, age, and ethnicity) could affect safety climate [7] and safety behavior [8], which are critical to worker safety. There are many examples of projects with poor safety performance due to managers ignoring the influence of demographic characteristics of workers. However, systematic studies about

AN INTEGRATED YOLO AND VLM SYSTEM FOR FIRE DETECTION IN ENCLOSED ENVIRONMENTS

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ABSTRACT

While YOLO models show promise in car fire detection, they remain insufficient for real-world deployment in underground and indoor car parks due to dataset limitations, evaluation gaps, and deployment constraints. We first fine-tune YOLO on fire/smoke-augmented dataset, but analysis reveals that it struggles with ambiguous fire-smoke boundaries, leading to false predictions. To address this, we propose a real-time end-to-end framework integrating YOLOv8s with Florence2 VLM, combining object detection with contextual reasoning. While YOLOv8s with VLM improves detection reliability, challenges are still ongoing. Our findings highlight YOLO's limitations in fire detection and the need for a more adaptive, environment-aware approach.

1 INTRODUCTION

A recent study (Meraner, 2023) highlights the growing risks of vehicle fires in underground/indoor car parks, emphasizing how the reduced spacing between adjacent cars -resulting from larger vehicle sizes and limited parking availability- leads to substantial property and environmental damage. Furthermore, Brzezinska & Bryant (2022) provides an in-depth review of the increasing fire incidents involving electric vehicles (EVs) powered by high-energy batteries, posing greater risks as the battery pack continues to burn even after water extinguishment, spreading to nearby vehicles within 5 minutes. Despite this fact, existing fire detection research (Moradi et al., 2024; Seydi et al., 2022; Shen et al., 2018; Gupta et al., 2021; Namozov & Im Cho, 2018; Son et al., 2018) primarily depends on multi-sensor information or focuses on detection in environments such as outdoor settings of wildland management to urban infrastructure (e.g., forest fires, building fires), or close-up fire imagery, leaving a critical gap in addressing vehicles fire detection in confine car parks. As a result, early fire suppression is crucial and only a few studies explored car fire detection in surveillance videos. Zhang et al. (2022) used a handmade dataset with a modified YOLOv4, while Dilshad et al. (2023) proposed an end-to-end system inspired by VGG16. However, both studies rely on custom datasets focused on already-developed flames, neglecting early smoke detection and indoor car park scenarios, where the risk of damage is highest.

First, we tune YOLOv8s capabilities by training it on our proposed dataset, handling the issue of data scarcity in vehicle fire detection within car park zones. Building on AI-generated data augmentation strategies (Rombach et al., 2022; Yang et al., 2023; He et al., 2022; Fang et al., 2024; Dunlap et al., 2023; Tian et al., 2024), we use Stable Diffusion (Rombach et al., 2022) to generate synthetic yet realistic images of vehicle fires set against car park backgrounds. This method pursues to address the lack of annotated fire and smoke images in these environments and provide a robust and diverse training dataset to effectively handle various real-world fire scenarios. However, through extensive analysis, we notice this is insufficient for immediate on-the-scene application.

To tackle the drawbacks of previous studies and handle car fire detection in confined parking zones, we further propose a novel End-to-End **(E2E)** fire detection framework specialized for challenging indoor and underground settings, by incorporating **VLM** (Vision-Language Model) as an auxiliary model to enhance the **YOLO** (You Only Look Once) (Redmon, 2016) object detection model. This unified approach seeks to adopt YOLO's real-time detection abilities and VLM's overall contextual scene understanding qualities.

In the end, we apply our framework to real-time CCTV footage of actual underground car parks. This comprehensive system processes live surveillance feeds, promptly detects fire or smoke with

*Equal contribution.

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Article

Effectiveness Analysis for Smart Construction Safety Technology (SCST) by Test Bed Operation on Small- and Medium-Sized Construction Sites

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Abstract: In the global construction industry, government policies have recently focused on smart construction technologies, such as those concerning the "smartization" of construction, improvements of productivity, and automation technologies. In addition, smart construction safety technologies (SCSTs) have been developed to ensure workers' safety, under the initiative of the private sector. In regards to overseas occupational safety, wearable technologies have been developed for various types of industries, and the integrated platform developments needed to link them have become mainstream. In South Korea, individual companies are focusing on developing basic SCSTs and platforms for integrated control, aiming to prevent accidents in the construction field. The goal of this study was to identify the pros and cons of SCSTs through test bed operation and to derive improvement directions. Therefore, a test bed embedded with SCSTs was built and operated to provide effective safety management for small- and medium-sized sites exposed to fatal accidents. From analyzing the data from the test bed, it was found that it is difficult to change the tendencies of workers' behaviors based solely on the introduction of SCSTs. This indicates that the effects of SCSTs are insignificant without the cooperation of workers. In addition, technical problems in field application were identified for each sensor and equipment, and the necessity, problems, and effectiveness of SCSTs were analyzed. As a result, both the installation and attachment types were found to be effective; however, workers avoided wearing certain attachment types. Based on the results derived through analysis of the pros and cons of SCSTs, the directions and guidelines were suggested for future use. This result can be used for future technology development directions, and policy establishment. Additionally, for the activation of SCSTs in the field, the cooperation of workers and the interest of managers remain essential factors, and improvements to the equipment are required.

Keywords: construction; automation technology; smart construction safety technology (SCST); safety management; small- and medium-sized site

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1. Introduction

Globally, changes have recently been demanded from the construction industry owing to social trends, such as aging workers, a lack of skilled manpower, and a reduction in working hours. The death toll from occupational accidents in construction is more than twice that from other industries; correspondingly, there is a need for the construction industry to solve its chronic safety problems [1]. According to the US Occupational Safety and Health Administration (OSHA), there were 1008 fatalities in the construction industry in 2020 (Figure 1a), accounting for approximately 20% of the total industry. Of these, 351 people died from fall accidents. OSHA also ranks the "Fatal Four" hazards accounting for the majority of injuries and deaths based on data from the U.S. Bureau of Labor Statistics, i.e., those accounting for over 60% of all construction worker deaths. These four major

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"Utilizing" Signal Detection Theory

Spencer K. Lynn and Lisa Feldman Barrett

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Decision-making and response strategies in interaction with alarms: the impact of alarm reliability, availability of alarm validity information and workload

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Enhancing workplace safety: A flexible approach for personal protective equipment monitoring

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ABSTRACT

Workplace safety is a prominent concern, motivating researchers across diverse disciplines to investigate valuable ways to address its challenges. However, creating an efficient system to address this issue remains a significant challenge. Since many accidents happen due to improper usage or complete removal of Personal Protective Equipment (PPE), one straightforward method for enhancing workplace security involves monitoring their usage This paper introduces an Operator Area Network (OAN) system which improves the existing solutions by increasing portability across different users and environments, non-intrusiveness and privacy. To enhance robustness in detecting the situations in which PPEs are not used correctly, we take advantage of Machine Learning to analyse the received signal strength indicator (RSSI) between PPEs in the same OAN The novelty of this work is that it does not exploit RSSI as a proxy of the distance but instead recognizes a signature of the correct wearing of the PPE By employing this system, employers can effectively ensure the proper usage of PPE devices at their worksites while also minimizing any adverse effects on workers' comfort and reducing the setup burden for employers. The system runs a Support Vector Machine (SVM) model several times per second and employs a post-processing algorithm to enhance its initial accuracy further As a result, the system effectively reduces false positives by about 80% and swiftly detects instances of improper usage of the worker's PPE, raising the alarm in less than seven seconds. Moreover, the post-processing algorithm can be customized to meet the specific needs of different use cases, allowing for a flexible trade-off between the detection time interval and the overall accuracy of the detection system.

1. Introduction

Ensuring workers' well-being, health and work performance requires appropriate working conditions and organizational structures (Davidescu et al., 2020). Specifically, safe and hygienic working conditions significantly impact both work quality and life outside the workplace (Greubel et al., 2016; Haar et al., 2014). Various legislation and initiatives reinforce these aspects, including the European Parliament Resolution of 10 March 2022 on a new EU strategic framework for health and safety at work after 2020. Despite these efforts, workplace safety remains a substantial challenge in modern society, as evidenced by statistics on workplace accidents (Eurostat, 2022). Furthermore, the study exploring disparities in working conditions among European Union member states, employing quantitative methodologies like TOPSIS and K-means (Tutak et al., 2022), underscores the need to focus on improving workplace safety. Addressing variations in working conditions, which encompass the effective utilization of PPE (Personal Protective Equipment), represents a fundamental element of workplace safety. Given the ongoing imperative to ensure workers' health and safety, a clear call exists to investigate novel approaches for monitoring PPE compliance. Embracing advanced monitoring technologies for PPE usage can offer valuable insights into safety protocol adherence, thus empowering organizations and policymakers to make informed decisions to elevate workplace safety standards and safeguard workers' well-being.

Workers are required to wear appropriate PPE as a first line of protection, in order to prevent workplace accidents from causing severe physical harm (Ammad, Alaloul, Saad, & Qureshi, 2021). Despite this, workers often do not comply with the use of PPE for various reasons, such as feeling uncomfortable during activities and lack of awareness (Li, Li, Luo, & Siebert, 2017; Wong, Man, & Chan, 2020).

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Calculation of signal detection theory measures

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Signal detection theory (SDT) may be applied to any area of psychology in which two different types of stimuli must be discriminated. We describe several of these areas and the advantages that can be realized through the application of SDT. Three of the most popular tasks used to study discriminability are then discussed, together with the measures that SDT prescribes for quantifying performance in these tasks. Mathematical formulae for the measures are presented, as are methods for calculating the measures with lookup tables, computer software specifically developed for SDT applications, and general purpose computer software (including spreadsheets and statistical analysis software).

Signal detection theory (SDT) is widely accepted by psychologists; the *Social Sciences Citation Index* cites over 2,000 references to an influential book by Green and Swets (1966) that describes SDT and its application to psychology. Even so, fewer than half of the studies to which SDT is applicable actually make use of the theory (Stanislaw & Todorov, 1992). One possible reason for this apparent underutilization of SDT is that relevant textbooks rarely describe the methods needed to implement the theory. A typical example is Goldstein's (1996) popular perception textbook, which concludes a nine-page description of SDT with the statement that measures prescribed by SDT "can be calculated . . . by means of a mathematical procedure we will not discuss here" (p. 594).

The failure of many authors to describe SDT's methods may have been acceptable when lengthy, specialized tables were required to implement the theory. Today, however, readily available computer software makes an SDT analysis no more difficult than a *t* test. The present paper attempts to demonstrate this and to render SDT available to a larger audience than currently seems to be the case.

We begin with a brief overview of SDT, including a description of its performance measures. We then present the formulae needed to calculate these measures. Next, we describe different methods for calculating SDT measures. Finally, we provide sample calculations so that readers can verify their understanding and implementation of the techniques.

We are indebted to James Thomas, Neil Macmillan, John Swets, Douglas Creelman, Scott Maxwell, Mark Frank, Helena Kadlec, and an anonymous reviewer for providing insightful comments on earlier versions of this manuscript. We also thank Mack Goldsmith for testing some of our spreadsheet commands. Correspondence concerning this article should be addressed to H. Stanislaw, Department of Psychology, California State University, Stanislaus, 801 West Monte Vista Avenue, Turlock, CA 95382 (e-mail: hstanisl@toto.csustan.edu).

OVERVIEW OF SIGNAL DETECTION THEORY

Proper application of SDT requires an understanding of the theory and the measures it prescribes. We present an overview of SDT here; for more extensive discussions, see Green and Swets (1966) or Macmillan and Creelman (1991). Readers who are already familiar with SDT may wish to skip this section.

SDT can be applied whenever two possible stimulus types must be discriminated. Psychologists first applied the theory in studies of perception, where subjects discriminated between *signals* (stimuli) and *noise* (no stimuli). The signal and noise labels remain, but SDT has since been applied in many other areas. Examples (and their corresponding signal and noise stimuli) include recognition memory (old and new items), lie detection (lies and truths), personnel selection (desirable and undesirable applicants), jury decision making (guilty and innocent defendants), medical diagnosis (diseased and well patients), industrial inspection (unacceptable and acceptable items), and information retrieval (relevant and irrelevant information; see also Hutchinson, 1981; Swets, 1973; and the extensive bibliographies compiled by Swets, 1988b, pp. 685–742). Performance in each of these areas may be studied with a variety of tasks. We deal here with three of the most popular: *yes/no* tasks, *rating* tasks, and *forced-choice* tasks.

Yes/No Tasks

A yes/no task involves *signal trials*, which present one or more signals, and *noise trials*, which present one or more noise stimuli. For example, yes/no tasks in auditory perception may present a tone during signal trials and nothing at all during noise trials, whereas yes/no tasks for memory may present *old* (previously studied) words during signal trials and *new* (distractor) words during noise trials. After each trial, the subjects indicate whether a sig-

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Editorial

Making sense of Cronbach's alpha

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International Journal of Medical Education

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Medical educators attempt to create reliable and valid tests and questionnaires in order to enhance the accuracy of their assessment and evaluations. Validity and reliability are two fundamental elements in the evaluation of a measurement instrument. Instruments can be conventional knowledge, skill or attitude tests, clinical simulations or survey questionnaires. Instruments can measure concepts, psychomotor skills or affective values. Validity is concerned with the extent to which an instrument measures what it is intended to measure. Reliability is concerned with the ability of an instrument to measure consistently.[1] It should be noted that the reliability of an instrument is closely associated with its validity. An instrument cannot be valid unless it is reliable. However, the reliability of an instrument does not depend on its validity.[2] It is possible to objectively measure the reliability of an instrument and in this paper we explain the meaning of Cronbach's alpha, the most widely used objective measure of reliability.

Calculating alpha has become common practice in medical education research when multiple-item measures of a concept or construct are employed. This is because it is easier to use in comparison to other estimates (e.g. test-retest reliability estimates)[3] as it only requires one test administration. However, in spite of the widespread use of alpha in the literature the meaning, proper use and interpretation of alpha is not clearly understood. [2, 4, 5] We feel it is important, therefore, to further explain the underlying assumptions behind alpha in order to promote its more effective use. It should be emphasised that the purpose of this brief overview is just to focus on Cronbach's alpha as an index of reliability. Alternative methods of measuring reliability based on other psychometric methods, such as generalisability theory or item-response theory, can be used for monitoring and improving the quality of OSCE examinations [6-10], but will not be discussed here.

What is Cronbach alpha?

Alpha was developed by Lee Cronbach in 1951[11] to provide a measure of the internal consistency of a test or scale; it is expressed as a number between 0 and 1. Internal consistency describes the extent to which all the items in a test measure the same concept or construct and hence it is connected to the inter-relatedness of the items within the test. Internal consistency should be determined before a test can be employed for research or examination purposes to ensure validity. In addition, reliability estimates show the amount of measurement error in a test. Put simply, this interpretation of reliability is the correlation of test with itself. Squaring this correlation and subtracting from 1.00 produces the index of measurement error. For example, if a test has a reliability of 0.80, there is 0.36 error variance (random error) in the scores (0.80×0.80 = 0.64; 1.00 – 0.64 = 0.36).[12] As the estimate of reliability increases, the fraction of a test score that is attributable to error will decrease.[2] It is of note that the reliability of a test reveals the effect of measurement error on the observed score of a student cohort rather than on an individual student. To calculate the effect of measurement error on the observed score of an individual student, the standard error of measurement must be calculated (SEM).[13]

If the items in a test are correlated to each other, the value of alpha is increased. However, a high coefficient alpha does not always mean a high degree of internal consistency. This is because alpha is also affected by the length of the test. If the test length is too short, the value of alpha is reduced.[2, 14] Thus, to increase alpha, more related items testing the same concept should be added to the test. It is also important to note that alpha is a property of the scores on a test from a specific sample of testees. Therefore investigators should not rely on published alpha estimates and should measure alpha each time the test is administered.[14]

Use of Cronbach's alpha

Improper use of alpha can lead to situations in which either a test or scale is wrongly discarded or the test is criticised for not generating trustworthy results. To avoid this situation an understanding of the associated concepts of internal consistency, homogeneity or unidimensionality can help to improve the use of alpha. Internal consistency is concerned with the interrelatedness of a sample of test items, whereas homogeneity refers to unidimensionality. A measure is said to be unidimensional if its items measure a single latent trait or construct. Internal consistency is a

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The Role of Automation in Right-time Construction Safety

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Abstract –

Construction organizations continue to be challenged to adequately prevent accidents. Although the five C's (culture, competency, communication, controls, and contractors) have been focusing for many years on compliance, good practices, and best in class strategies, even industry leaders have only marginal improvements in recorded safety statistics. Right-time vs. real-time construction safety and health identifies three major focus areas to aid in the development of a strategic – as opposed to a tactical – response. (1) Occupational safety and health by design, (2) real-time safety and health monitoring and alerts, and (3) education, training, and feedback leveraging state-of-the-art technology provide meaningful predictive, quantitative, and qualitative measures to identify, correlate, and eliminate hazards before workers get injured or incidents cause collateral damage. Based on the current state-of-the-art of existing innovative initiatives in the occupational construction safety and health domain, a framework for right-time vs. real-time construction safety and health presents the specific focus on assisted safety and health data gathering, analysis and reporting to achieve a better safety performance. The practical as well as social implications in conducting a rigorous right-time safety culture in a construction business and its entire supply chain are tested in selected application scenarios and results are presented.

Keywords –

Right-time alert and warning technologies, automation and control, construction safety and health, historical safety statistics, lagging and leading indicators, performance measures, prevention through design, pro-active decision making, real-time monitoring and feedback, remote sensing, safety culture and climate, worker education and learning.

1 Introduction

Many safety studies that were led by industry or research experts focused greatly on traditional safety assessment and improvement techniques. In fact, these efforts have been successful and have changed the culture of safety. Today, many owners take safety very seriously and demand from contractors or subcontractors – and slowly from the entire supply chain in construction – to provide excellent track records and safety management on their jobsites. But even voluntarily reported safety performance statistics by industry leaders show that the real impact of safety best practices – which can be much stricter than existing federal or state safety rules regulations – hardly leads to further decreases in recordable incident rates. A valid question remains: How can the goal of zero accidents in construction and in any other hazardous workplace environments be achieved?

To review the magnitude of this problem, it is critical to consider construction with its significance to the economy of many countries. The U.S. construction industry, for example, accounts for $611 billion or 4% of the yearly gross domestic product (10% if equipment, furnishings, energy and other sectors are included), employs 7% of the total workforce, and invests $1.16 trillion every year in the economy [1]. A key message is that advances in emerging technologies offer significant opportunities to improve construction safety substantially and help meet other national challenges, such as becoming more efficient in work tasks. Thus, an investment in safety directly impacts the performance of other troubled industry sectors.

A key element that can help in objective data collection, processing, information, and knowledge generation of many of the mostly dynamic construction resources (personnel, equipment, and materials) [2] is automation [3]. Finding ways to achieve such automation and leveraging information that becomes available at the right-time vs. real-time is the topic of this paper. The author argues that such automation should be done at the right-time, enabling up-to-date safety data gathering, analysis and reporting when needed and with the ultimate goal of achieving better safety performance.

The paper outline has the following structure: the background reviews occupational safety and health literature; second, up-to-date safety performance statistics and performance measurement techniques are reviewed; third, the motivation for right-time vs. real-



Probable Inference, the Law of Succession, and Statistical Inference

Edwin B. Wilson

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MonitorVLM: A Vision–Language Framework for Safety Violation Detection in Mining Operations

Jiang Wu#, Sichao Wu#, Yinsong Ma, Guangyuan Yu, Haoyuan Xu, Lifang Zheng, Jingliang Duan*

***Abstract*—Industrial accidents, particularly in high-risk domains such as surface and underground mining, are frequently caused by unsafe worker behaviors. Traditional manual inspection remains labor-intensive, error-prone, and insufficient for large-scale, dynamic environments, highlighting the urgent need for intelligent and automated safety monitoring. In this paper, we present MonitorVLM, a novel vision–language framework designed to detect safety violations directly from surveillance video streams. MonitorVLM introduces three key innovations: (1) a domain-specific violation dataset comprising 9,000 vision–question–answer (VQA) samples across 40 high-frequency mining regulations, enriched with augmentation and auxiliary detection cues; (2) a clause filter (CF) module that dynamically selects the Top-K most relevant clauses, reducing inference latency by 13.56% while maintaining accuracy; and (3) a behavior magnifier (BM) module that enhances worker regions to improve fine-grained action recognition, yielding additional gains of 3.45% in precision and 8.62% in recall. Experimental results demonstrate that MonitorVLM significantly outperforms baseline vision–language models, achieving improvements of 22.01% in precision, 34.22% in recall, and 28.37% in F1 score over the 72B unfine-tuned baseline. A lightweight web-based interface further integrates MonitorVLM into practical workflows, enabling automatic violation reporting with video timestamping. This study highlights the potential of multimodal large models to enhance occupational safety monitoring in mining and beyond.**

***Index Terms*—Industrial safety monitoring, Vision-language model, Mining operations, Violation detection**

I. Introduction

The vast majority of industrial accidents, including those occurring in mining and construction, originate from unsafe worker behaviors, which highlights the urgent need for continuous monitoring and timely early-warning systems [1]. In practice, safety management on industrial sites still depends heavily on manual inspections conducted by human supervisors. Such inspections are not only labor-intensive and time-consuming, but also prone to human error, inconsistency, and omissions when faced with the scale and complexity of modern worksites [2]. As industrial projects grow larger and more dynamic, these limitations make purely manual inspection increasingly inadequate for ensuring effective and reliable safety oversight.

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Given the limitations of manual inspection, researchers have increasingly explored automated approaches for safety violation detection. Existing methods can be broadly categorized into three groups: (1) object-detection models, which rely on bounding-box localization to identify unsafe objects or worker postures; (2) zero-shot semantic models, which embed images and textual prompts into a shared space to match visual content with safety clauses; and (3) vision–language models (VLMs), which jointly process multimodal inputs to generate natural-language violation descriptions aligned with regulations.

Early automated safety monitoring efforts largely adapted object-detection networks to industrial scenarios, with a primary emphasis on personal protective equipment (PPE) detection. Fang et al. [3] trained Faster R-CNN to locate workers' heads and classify them as "hard-hat" or "no-hard-hat", enabling automated helmet compliance checks. Building on this idea, Imam et al. [4] extracted human keypoints to compute joint angles and incorporated relative body size (normalized by bounding-box height) as an auxiliary feature. These signals were then integrated into the YOLO-Pose family (v8–v11, YOLO-World, and RT-DETR) for end-to-end training, allowing real-time identification of hazardous worker postures. Similarly, Vukicevic et al. [5] employed the open-source HigherHRNet [6] to precisely localize five body regions, including head, eyes, hands, torso, and feet, and combined this with MobileNetV2 to verify whether each of 18 custom safety items was worn in the correct location. Although these methods deliver high accuracy in PPE violation detection, their scope remains narrow: object detectors cannot capture contextual violations, reason about sequential actions, or interpret the deeper semantics of complex site conditions [7].

The advent of zero-shot semantic models [8] introduced a new paradigm for industrial safety inspection, enabling images and text to be embedded in a shared feature space so that regulatory clauses could be matched directly with visual content. By operating at the semantic level, CLIP [9] substantially improves both the flexibility and interpretability of violation detection. Gil [10] demonstrated this capability by providing only two prompts, "compliant/violation," allowing CLIP to score entire images without additional training or explicit annotations of PPE locations, and thus identify images containing potential hazards. Building on this idea, Hsiao et al. [11] prefix-tuned CLIP on mining-specific image–text pairs and then fed the aligned features into a lightweight captioning module that generated natural-language descriptions with keywords such as "no helmet" or "materials piled dangerously." These captions enabled zero-shot hazard recognition and automatic incident reporting in text form. More recently, Tsai et

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ConstructAI: From Real-Time Safety Insight to Skill Growth in Deployed Construction AI Systems

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Abstract

Ensuring safety in power grid construction remains a critical yet challenging task, as existing monitoring approaches often lack scalability, timeliness, and adaptability to diverse on-site conditions. To address these limitations, we present **ConstructAI**, a deployed AI-driven safety management system that integrates multi-source image and video acquisition devices with advanced multimodal large model reasoning. The system combines text, image, and video prompts through an efficient workflow powered by LLaMA3 and Meta SAM2 backbones, enhanced with LoRA and adaptor modules for multimodal fusion. Once deployed, ConstructAI continuously processes real-time construction footage to identify violations, assess risk levels, and generate standardized rectification requirements. The deployment has demonstrated measurable benefits across multiple sites, including a 70% increase in violation rectification rates, reduction of average rectification delays from hours to minutes, and a 45% decline in repeat violations. Beyond technical gains, ConstructAI has delivered significant business impacts, such as reduced safety incidents, improved compliance with national regulations, and higher operational efficiency. By enabling proactive risk management and structured safety feedback loops, our system exemplifies how innovative use of AI can translate into tangible improvements for industrial safety. The lessons learned from deployment highlight the importance of balancing algorithmic advances with practical integration into organizational workflows.

Introduction

Ensuring safety and improving workforce development in construction sites remain global challenges. Construction is consistently ranked among the most hazardous industries worldwide, with high rates of workplace accidents and injuries (Wadsworth and Walters 2019). Workers are frequently exposed to tasks such as working at height, handling heavy materials, and operating complex machinery in rapidly changing environments. Many of them join the workforce with limited formal training, which exacerbates safety risks and contributes to both acute incidents and chronic physical strain (Zhou, Whyte, and Sacks 2012). These challenges not only affect individual well-being but also impose significant economic costs on the industry (Brynjolfsson and Mcafee 2017).

Traditional safety management approaches—manual supervision, periodic inspections, and classroom-based training—are insufficient in large and complex projects. Supervisors cannot continuously monitor all workers across dispersed sites, while training content is often forgotten or inconsistently applied once workers are on the job (Teizer 2016). As a result, unsafe behaviors are often addressed only after incidents occur, and opportunities for proactive prevention and continuous skill growth are missed (Fang et al. 2020).

To address these limitations, we developed ConstructAI, a deployed AI system for real-time safety compliance monitoring and worker skill development in construction. The system continuously analyzes video streams captured from helmet-mounted first-person cameras, fixed-site surveillance, and drone inspections. By integrating egocentric action recognition, cross-view video fusion, and rule-informed compliance reasoning (Chen et al. 2024), ConstructAI detects non-compliant behaviors and physical strain risks in real time and pushes corrective feedback directly to workers' mobile devices.

Since early 2024, ConstructAI has been deployed across more than a dozen large-scale construction projects in China. In one deployment phase, the system automatically recorded 929 safety violation cases involving 168 workers across projects in Suzhou, Shenzhen, and Changzhou. Each violation entry was linked to first-person evidence video, a corrective action requirement, and rectification confirmation, forming a complete compliance cycle. Overall, deployment results demonstrated substantial impact. Safety rectification rate increased from 2.3% to 75.8%, Average response time dropped from 6 hours to $<$10 seconds. Repeat violations decreased by 45%. Minor injury incidents were reduced by 36%, and over 9,000 individualized feedback reports were generated to support workforce training.

The contributions of this paper are threefold:

- We design an automated annotation pipeline to generate fine-grained referring expressions for construction safety

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산업안전보건법

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제1장 총칙

제1조(목적) 이 법은 산업 안전 및 보건에 관한 기준을 확립하고 그 책임의 소재를 명확하게 하여 산업재해를 예방하고 쾌적한 작업환경을 조성함으로써 노무를 제공하는 사람의 안전 및 보건을 유지·증진함을 목적으로 한다. <개정 2020. 5. 26.>

제2조(정의) 이 법에서 사용하는 용어의 뜻은 다음과 같다. <개정 2020. 5. 26., 2023. 8. 8.>

1. "산업재해"란 노무를 제공하는 사람이 업무에 관계되는 건설물·설비·원재료·가스·증기·분진 등에 의하거나 작업 또는 그 밖의 업무로 인하여 사망 또는 부상하거나 질병에 걸리는 것을 말한다.
2. "중대재해"란 산업재해 중 사망 등 재해 정도가 심하거나 다수의 재해자가 발생한 경우로서 고용노동부령으로 정하는 재해를 말한다.
3. "근로자"란 「근로기준법」 제2조제1항제1호에 따른 근로자를 말한다.
4. "사업주"란 근로자를 사용하여 사업을 하는 자를 말한다.
5. "근로자대표"란 근로자의 과반수로 조직된 노동조합이 있는 경우에는 그 노동조합을, 근로자의 과반수로 조직된 노동조합이 없는 경우에는 근로자의 과반수를 대표하는 자를 말한다.
6. "도급"이란 명칭에 관계없이 물건의 제조·건설·수리 또는 서비스의 제공, 그 밖의 업무를 타인에게 맡기는 계약을 말한다.
7. "도급인"이란 물건의 제조·건설·수리 또는 서비스의 제공, 그 밖의 업무를 도급하는 사업주를 말한다. 다만, 건설공사발주자는 제외한다.
8. "수급인"이란 도급인으로부터 물건의 제조·건설·수리 또는 서비스의 제공, 그 밖의 업무를 도급받은 사업주를 말한다.
9. "관계수급인"이란 도급이 여러 단계에 걸쳐 체결된 경우에 각 단계별로 도급받은 사업주 전부를 말한다.
10. "건설공사발주자"란 건설공사를 도급하는 자로서 건설공사의 시공을 주도하여 총괄·관리하지 아니하는 자를 말한다. 다만, 도급받은 건설공사를 다시 도급하는 자는 제외한다.
11. "건설공사"란 다음 각 목의 어느 하나에 해당하는 공사를 말한다.
    가. 「건설산업기본법」 제2조제4호에 따른 건설공사
    나. 「전기공사업법」 제2조제1호에 따른 전기공사
    다. 「정보통신공사업법」 제2조제2호에 따른 정보통신공사
    라. 「소방시설공사업법」에 따른 소방시설공사
    마. 「국가유산수리 등에 관한 법률」에 따른 국가유산 수리공사
12. "안전보건진단"이란 산업재해를 예방하기 위하여 잠재적 위험성을 발견하고 그 개선대책을 수립할 목적으로 조사·평가하는 것을 말한다.

BOSCH

Alarm Verification for Alarm Management



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